Cryptography

Unit- 2.2
Modern Ciphers

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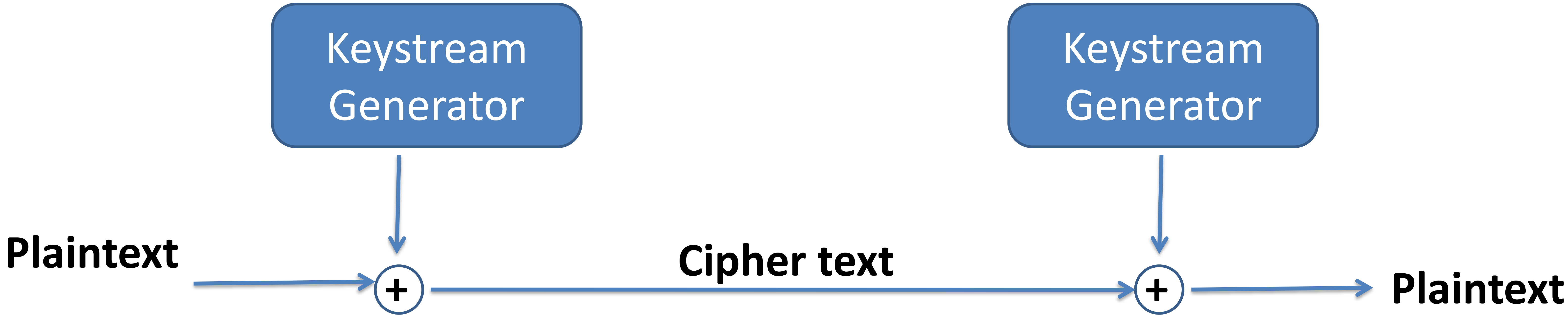
Stream Cipher

- Stream cipher is a type of encryption algorithm that operates on individual bits or bytes of data, and encrypts or decrypts data in a continuous stream.
- In stream cipher, a secret key is used to generate a key stream of random bits, which is then combined with the plaintext using bitwise X-OR operation to produce the cipher text.



Stream Cipher

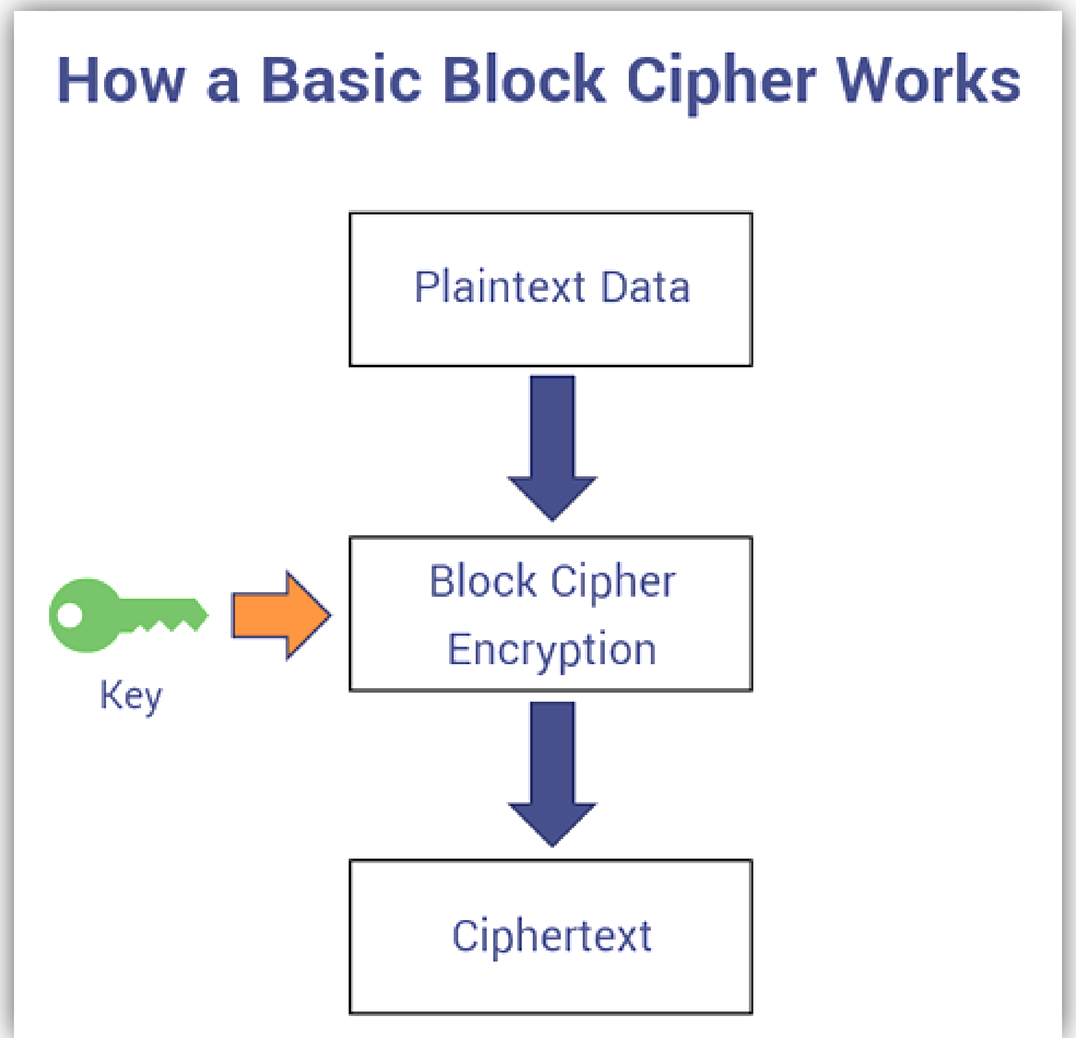
- Unlike block cipher, which encrypt data in a fixed size blocks, stream ciphers encrypt data in a continuous stream of data, typically one byte or bit at a time.

- One of the advantage of stream ciphers is that they can be implemented with relatively low computational overhead, making them well suited for applications where high performance is important.

- Example:
  1. FISH
  2. ISAAC
  3. RC4
  4. SEAL
  5. SNOW etc.

Block Cipher

- Block cipher is a type of encryption algorithm that operates on fixed-size blocks of data, typically 64 or 128 bit in length.
- In block cipher, a secret key is used to transform a plaintext block in to a corresponding ciphertext block and the same key is used to decrypt the ciphertext back in to plaintext.
- Block ciphers are typically implemented as a sequence of encryption rounds, each of which involves applying a permutation or substitution function to the input block using a key dependent transformation.
- The output of each round is then fed into the next round as input creating a chain of transformations that ultimately produce the final ciphertext block.
- Examples: DES, AES, IDEA, RC5, BlowFish etc.



Block Vs Stream Cipher

Block Cipher	Stream Cipher
1. Encoding of the plaintext is done as a fixed length block one by one. A block for example could be 64 or 128 bit.	1. Encoding of plaintext is done bit by bit or byte by byte.
2. Same key is used multiple time.	2. Same key is used but only for one encryption.
3. Organized as a sequence of round where each round's out put is the input for the next round.	3. Key is used to generate key space and then key space are X-ORed with the plaintext.
4. Large key size.	4. Small key size.
5. More secure in most cases.	5. Equally secure if properly designed.
6. More complex and slower.	6. Usually very simple and much faster.
7. Eg, DES, AES, IDEA, RC5	7. Eg, FISH, ISSAC, RC4

Symmetric Vs Asymmetric Cipher

Symmetric Cipher	Asymmetric Cipher
1. Same key is used for encryption and decryption.	1. Different key is used for encryption and decryption i.e. public key for encryption and private key for decryption.
2. Shared secret key must be securely distributed among all communicating parties.	2. Public key can be openly shared, while private keys must be kept secret.
3. Number of keys required increases exponentially with the number of communication parties.	3. Scalable, as each user has their own key pair.
4. No inherent authentication of communicating parties.	4. Provide inherent authentication and digital signatures.
5. No forward secrecy, as compromise of shared key compromises past and future communications .	5. Provide forward secrecy, as each key pair is independent.
6. Generally faster, algorithms are more efficient..	6. Slower, algorithms are typically expensive.
7. Eg, DES, AES, 3DES	7. Eg RSA, DSA, ECC

Cryptanalysis

- In simple word, cryptanalysis is a technique of decoding messages from a non-readable format to a readable format with out knowing a key.
- In technical term, cryptanalysis is the practice of studying and breaking cryptographic systems in order to gain unauthorized access to encrypted data.
- Cryptanalysis techniques vary depending on the type of encryption algorithm being used, but generally involve using statistical analysis, algebraic equations, and computational methods to deduce the key or plaintext.
- To defend against the cryptanalysis attack, it is important to use strong encryption algorithms, choose long and complex keys and follow best practices for key management and data protection.

Cryptology

- Cryptology is the study of cryptography and cryptanalysis, the practice of creating and breaking codes and ciphers to protect or obtain information.

Cryptography

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Cryptanalysis

Cryptology

Cryptanalysis Attack Model

- The various cryptanalytic attack model are as follows-

1. Ciphertext only attack:

- Ciphertext only attack is a type of cryptanalysis attack where the attacker only has access to the ciphertext of an encrypted message and does not have any knowledge of the plaintext or the encryption key used to encrypt the message.

- Here the attacker's goal is to recover the plaintext or the encryption key or to gain other information about the encrypted message. To do this attacker typically uses statistical analysis or other techniques to analyze the ciphertext and look for patterns or characteristics that can provide clues about the plaintext or the key.

- Most difficult type of attack.

2. Known plaintext attack:

- Known plaintext attack is a type of cryptanalysis attack where the attacker has access to both the ciphertext and some corresponding plaintext of the same message.
- This type of attack assumes that the attacker has some knowledge of the encryption algorithm or system used to encrypt the message.

- Known plaintext attacks can be more powerful than ciphertext only attack, as they provide additional information to the attacker. However, they are still considered difficult type of attack, especially if the encryption algorithm is strong and the key length is sufficiently long.

3. Chosen plaintext attack:

- Chosen plaintext attack is a type of cryptanalysis attack where the attacker has the ability to choose and encrypt plaintext messages of their choice using the encryption system being analyzed. This type of attack assumes that the attacker has some knowledge of the encryption algorithm or system used to encrypt the messages.

- Here the attackers goal is to recover the encryption key or gain additional information about the encryption system by analyzing the chosen plaintext and the corresponding ciphertext.

- Chosen plaintext attack are more powerful than known plaintext attacks or ciphertext only attacks.

4. Chosen ciphertext attack:

- A choosen ciphertext attack is a type of cryptanalysis attack where the attacker has the ability to choose and decrypt ciphertext message of their choice using the decryption system being analyzed.

4. Brute Force Attack

- Try all possible keys until correct one is found.

!!!Thank you!!!